

Person Re-Identification Using Vision Transformers and Efficient CNNs

I have documented my work in detail for both the ViT pipeline and the ResNet pipeline.

[Link 1: CNN](#) (Please change the document tabs on the left.)

[Link 2: ViT](#)

Problem Statement

The goal of this project is to design a robust person re-identification system that can reliably match individuals across different camera views while handling challenges such as viewpoint changes, occlusion, and visually similar clothing. The system should learn discriminative identity features that generalize across cameras and datasets.

1. Introduction

Person Re-Identification (Re-ID) aims to match the same individual across multiple non-overlapping camera views. This task is challenging because of variations in lighting, camera viewpoint, pose, occlusion, and background clutter.

In this project, I made a Vision Transformer (ViT) based baseline model for person Re-ID on the Market-1501 dataset. After evaluating the baseline model, I conducted a detailed error analysis to identify its main limitations, particularly problems in data and cross-camera appearance variations. Based on these observations, I developed an improved ViT V2 architecture incorporating camera-aware embeddings and improved training strategies. In addition, to explore deployment efficiency, I built a lightweight ResNet50-based model optimized for faster inference and edge deployment.

2. Dataset

The model is trained only on Market-1501 and tested directly on DukeMTMC (For Cross-Dataset Test) without any fine-tuning to measure its ability to generalize to unseen environments and camera setups.

Table 1: Market-1501 Dataset Statistics

Part	Images	Identities	Cameras
Training set	12,936	751 persons	6 cameras
Query set	3,368	750 persons	6 cameras
Gallery set	19,732	751 persons	6 cameras
Total images	32,668	1,501 persons	6 cameras

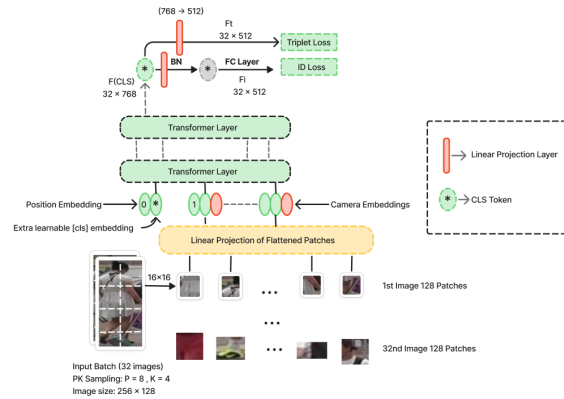
Table 2: DukeMTMC-reID Dataset Statistics

Part	Images	Identities	Cameras
Training set	16,522	702 persons	8 cameras
Query set	2,228	702 persons	8 cameras
Gallery set	17,661	1,110 persons	8 cameras
Total images	36,411	1,404 persons	8 cameras

3. Methodology Overview

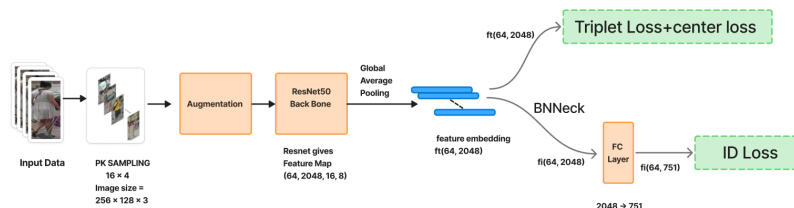
To explore both accuracy and deployment efficiency, two complementary architectures were used. The first model is a DeiT-Base (ViT) based Re-ID system designed to learn strong global identity representations with camera embeddings. The second model is a ResNet50-based pipeline designed to provide competitive accuracy while remaining lightweight and suitable for edge deployment.

3.1 Vision Transformer Pipeline (ViT)



The V2 (With camera embeddings, ViT.) architecture was built after V1(Without camera embeddings, ViT.) error analysis revealed the model was struggling with cross camera appearance variation due to missing camera embeddings and label smoothing. Training uses PK sampling ($P=8$, $K=4$) forming batches of 32 images of size 256×128 . Each image is divided into 128 non-overlapping 16×16 patches and linearly projected to 768-dim tokens. A learnable CLS token is prepended, position embeddings are added to all tokens, and Camera SIE ($\lambda=0.1$) is injected into patch tokens Z1–Z128 only. The token sequence passes through 12 DeiT-Base transformer layers and the output CLS token F(CLS) serves as the global identity representation, projected to 512-dim via BNNeck producing Ft (pre-BN) and Fi (post-BN). Ft is used for Batch-Hard Triplet Loss and Fi for ID Loss with label smoothing ($\epsilon=0.1$), both optimised jointly using Adam with 10-epoch warmup and cosine decay. Dual learning rates are applied backbone at $0.1 \times$ to preserve pretrained weights, head at $1.0 \times$ for faster convergence. Each image is passed twice (original + flipped), embeddings averaged and L2 normalised (Flip TTA) for query. K-reciprocal re-ranking ($k1=20$, $k2=6$, $\lambda=0.3$) is applied as post-processing, replacing cosine distance with Jaccard distance to correct ranking errors without retraining.

3.2 ResNet50 Pipeline



Training Pipeline: Input images are sampled using PK sampling ($P=16$ identities $\times$ $K=4$ images, batch size=64) and passed through augmentations including random flip, random crop, and random erasing. The ResNet50 backbone (last stride=1) processes each image and outputs a spatial feature map of shape (64, 2048, 16, 8). Global Average Pooling compresses this into a 2048-dimensional vector ft . **Embeddings and Losses:** ft is the raw feature embedding used directly for Triplet Loss (batch-hard mining) and Center Loss ($\beta=0.0005$) these two losses pull same-identity embeddings closer and push different identities apart in feature space. ft then passes through BNNeck (a single BatchNorm layer) to produce the normalized embedding fi . fi goes into the FC classifier (2048 $\rightarrow$ 751) for ID Loss (cross-entropy + label smoothing $\epsilon=0.1$). **Inference:** The FC classifier is discarded. Query and gallery images are passed through backbone $\rightarrow$ GAP $\rightarrow$ BNNeck to extract fi . fi is L2-normalized and matched against gallery embeddings using cosine distance, refined with k-reciprocal re-ranking ($k_1=20$, $k_2=6$, $\lambda=0.3$).

4. Results

Table 3: Comparison of ViT and ResNet50 Models with and without Re-ranking

Model	Rank-1	Rank-5	Rank-10	mAP
ViT V1 (Baseline)	86.79%	–	–	69.71%
ViT V1 + Re-ranking	89.07%	–	–	83.16%
ViT V2 (Final Model)	88.92%	–	–	73.07%
ViT V2 + Re-ranking	91.83%	–	–	86.39%
ResNet50	94.30%	98.37%	99.08%	85.70%
ResNet50 + Re-ranking	95.46%	97.80%	98.34%	94.08%

The models were evaluated on the Market-1501 benchmark using standard person re-identification metrics including Rank-1, Rank-5, Rank-10 accuracy and mean Average Precision (mAP). Results are reported both before and after applying k-reciprocal re-ranking.

Table 4: Model Efficiency Comparison

Model	Parameters	Model Size	Single Image Latency	Throughput
ViT V2	86.5M	330 MB	10.92 ms	164 images/sec
ResNet50 (FP32)	25M	95.56 MB	5.47 ms	$\sim$ 425 images/sec
ResNet50 (FP16)	25M	47.78 MB	$\sim$2.8 ms	$\sim$440 images/sec

Mixed precision inference significantly improves deployment efficiency. Converting the ResNet50 model to FP16 reduces the model size by 50% and nearly doubles inference speed while maintaining identical retrieval accuracy.

Table 5: Cross-Dataset Generalization (Market-1501 → DukeMTMC-reID)

Model	Training Dataset	Testing Dataset	Rank-1	mAP
ViT V2 (Baseline)	Market-1501	DukeMTMC-reID	38.24%	24.98%
ResNet50 (Baseline)	Market-1501	DukeMTMC-reID	26.17%	13.69%

After applying post-processing techniques such as AQE and DBA, the Rank-1 accuracy of the ViT model improved to 42.32%, indicating that the learned embedding space is well-structured and capable of grouping similar identities effectively. ResNet with center embeddings gave Rank-1 71.10 %

5. Error Analysis and Limitations

I did an in-depth error analysis starting from the baseline model I created, which was the ViT V1 baseline. From this, I found a data insight that there were some problems in the data that could not be discovered just from EDA. The problems were that in some PIDs in the query and gallery, there were two different persons in the same PID, and in another case, the same person was given two PIDs. These issues cause the model to penalize wrongly even when the model prediction is correct. Secondly, I found that V1 lacked in capturing camera angles, so a few improvements were made in V2. From the error analysis, I also found that it lacked in capturing local features. For that, I thought of creating a local feature embedding, which can be my future work. In ResNet, I found that the model was good in many cases but fails in tough occlusion cases, so for that also we need a strong ViT-based pipeline. Also, from cross-domain error analysis, I found that ViT has the capability of generalizing well compared to the ResNet pipeline, which works well on the same data and the same test set.

6. Conclusion

The results show that robust identity embeddings can be learned using both transformer and CNN architectures. While the ViT model captures strong global representations, the ResNet50 pipeline offers an efficient and deployable solution.

7. Code

I did the complete model development on Kaggle T4 GPU. For your convenience, you can look at my Kaggle notebooks, which contain both the code and the output logs. (Model.pth are saved)

DeiT-Base (ViT) Code : V1 Model → ViT Baseline V2 Model → ViT With camera Embed.

V1 Training and Inference: [Link](#) (Check version 5.)

V2 Training: [Link](#)

V2 Inference: [Link](#)

ResNet50 Pipeline Notebooks:

Training: [Link](#) Inference: [Link](#)

8. Checkpoints (best model weights)

The checkpoints of the models are available in the zip folder at the provided link. It contains the V1 model weights, V2 best model weights, and the ResNet model weights.

[Link to the folder.](#)